

MAE 3780 Mechatronics, Robot Project Final Report

Group 18

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1. Robot Design and Strategy Overview

Our robot was built on the stock kit chassis with two modified Parallax continuous rotation servos, each driven by an L9110H H-bridge. We added two 3D-printed extensions on either side to widen the robots perimeter and sweep cubes inward. For sensors we used a TCS3200 color sensor on the bottom of the chassis and two QTI sensors at the front left and right edges for border detection, all connected to an Arduino UNO r3. All code was written in C using AVR registers.

The basic idea was to sweep back and forth across the field and collect cubes as we went. At the start of each match the robot took ten readings from the TCS3200 to establish a baseline for whichever color zone it started on. It then drove forward until the sensor detected a period change greater than 150 units from that baseline, which meant it had crossed into the other color zone. At that point it nudged forward, turned left or right (alternating each crossing), moved over slightly, updated its color baseline, and kept going, producing a back-and-forth path across the field. The QTI sensors detected the border, if one sensor went over the black border it triggered a corrective turn, and if both hit black at the same time it did a longer right turn to get back onto the field.

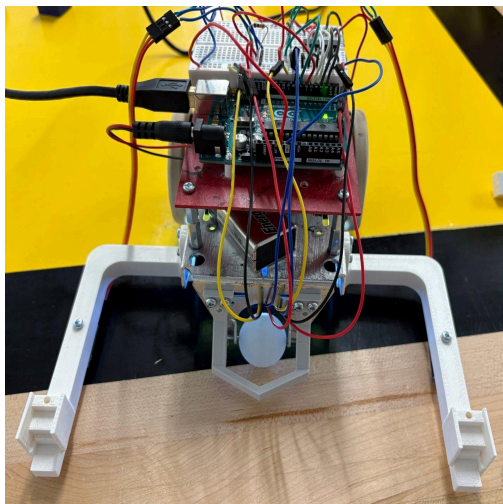


Figure 1. Competition day robot with fixed 3D-printed extensions.

2. Design Process Reflection

We originally planned two servo-actuated arms that could open to gather cubes and close to hold them. We got pretty far into writing the servo control code, PWM setup via Timer 1 and a `setServoAngle()` function, before realizing we had a size problem. That code is still visible in commented-out sections of the final submission.

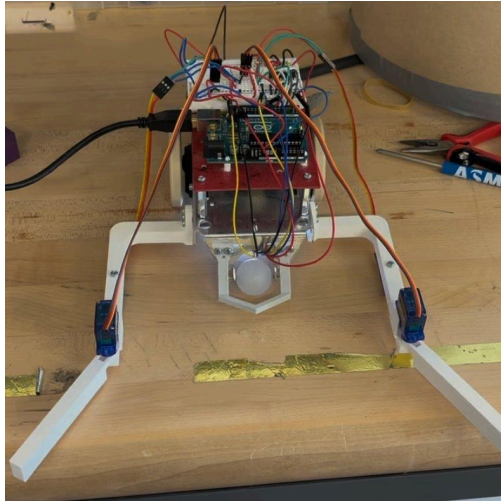


Figure 2. Original design with servo-actuated arms open.

When we put the robot in the size box late in the build period, the open arms pushed it past the 12-inch diameter cylinder limit. Without enough time to redesign from scratch, we got rid of the extension arms and just kept the fixed extension arms and bolted them to the chassis. It wasn't what we wanted, but it got us within the constraint and we kept the width.

Milestone 2 got us to get good drive motion early, which everything else built on. Milestone 3 took the most time. We spent a lot of time tuning the sensor thresholds because the TCS3200 and QTI readings shifted a lot depending on lighting and where we tested, and we were originally only using one QTI which made border detection unreliable. Milestone 4 confirmed the sweep approach actually worked before we committed to it for competition.

3. Competition Analysis

We won our round robin group and advanced to single elimination. The sweep worked pretty reliably, especially against smaller robots that left gaps near the border. One thing we didn't anticipate was that the rigidity of the extensions let us

knock opponents off the board during contact, which caused them to lose cubes they had already gathered. That wasn't part of the plan but it helped.

In the bracket we won the first match but lost in the second round to a robot that was basically just a large block filling most of its allowed footprint. We couldn't sweep around it, every time we made contact we lost our position and had to reorient. It also just sat on cubes, and there wasn't much we could do about that with a sweep strategy.

Speed was another issue we noticed during the bracket. Since we only had two drive motors and passive sensors, we had extra power available that never went anywhere. A faster robot would have gotten more sweeps in per minute and recovered quicker after corrections.

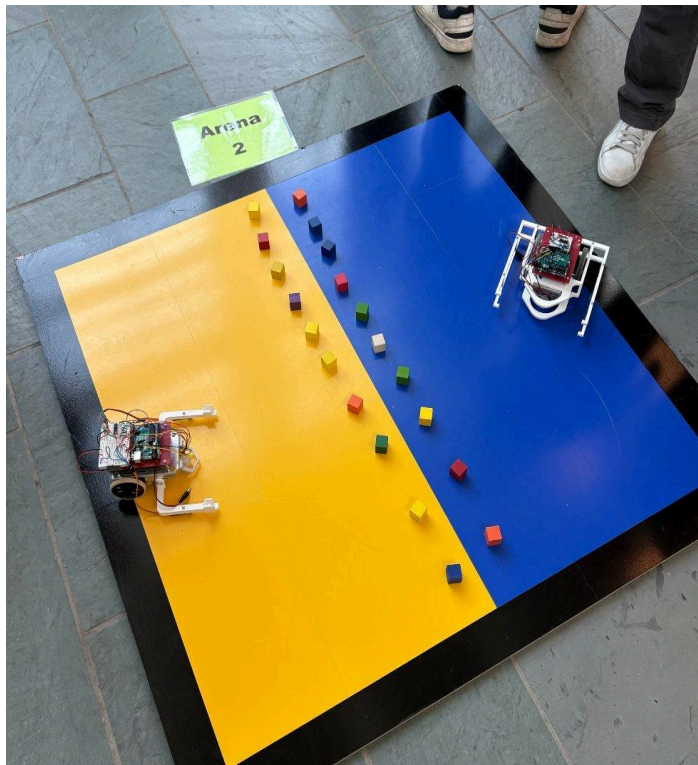


Figure 3. Our robot (yellow side) during a match in the Atrium.

4. Conclusions

We made it out of round robin which was really exciting, but losing in the bracket showed we still had two things we hadn't really figured out, what to do against a robot you can't get around, and how to stop losing cubes we had already gathered.

If we did this again the biggest thing would be designing the servo arms properly from the start instead of scrambling at the end. We knew we wanted closing arms, we just ran out of time to make them fit.

For future groups, check your dimensions against the size box earlier in the process. Also, the lighting in Duffield is noticeably different from the lab, test your sensor thresholds there at least once before competition day.

Appendix A

Bill of Materials						
Name	Part #	Description	Supplier	Quantity	Unit Price (\$)	Total Cost (\$)
Left Arm	001	ABS 3D-printed left collection arm rigidly attached to robot chassis.	RPL	1	9.97	9.97
Right Arm	002	ABS 3D-printed right collection arm rigidly attached to robot chassis.	RPL	1	9.97	9.97
Left Gate	003	ABS 3D-printed swing gate to extend the robot arm reach.	RPL	1	4.54	4.54
Right Gate	004	ABS 3D-printed swing gate to extend the robot arm reach.	RPL	1	4.54	4.54
Duct Tape	005	Adhesive for part mounting.	Walmart	1	5.77	5.77
Front Guard	006	PLA 3D-printed guard for front wheel.	RPL	1	2.70	2.70
Total:				6		\$37.49

Appendix B

Circuit Diagram

